HDOC Day-68

Recursive Conversion of Decimal Number to Binary

Problem Statement

Write a recursive function to convert a decimal number into its binary representation. The program accepts a non-negative decimal integer and displays its binary digits using recursion.

Algorithm

1. Start.

2. Read a non-negative decimal integer n.

3. Call decimal_to_binary(n).

4. In the recursive function, if n > 1, first call decimal_to_binary(n / 2).

5. Print n % 2 after the recursive call returns. This produces binary digits in the correct left-to-right order.

6. Print a newline and stop.

Test Case Table

Test Case	Decimal Input	Expected Binary Output	Purpose
1	0	0	Zero / base case
2	1	1	Single binary digit
3	2	10	Small even number
4	5	101	Odd number
5	10	1010	Even number
6	13	1101	Typical input
7	25	11001	Larger odd number
8	42	101010	Alternating digits
9	255	11111111	Eight one-bits
10	1024	10000000000	Power of two

C Program

```
#include <stdio.h>

void decimal_to_binary(unsigned int n)
{
    if (n > 1)
        decimal_to_binary(n / 2);
    printf("%u", n % 2);
}

int main(void)
```

```
{
    unsigned int n;
    printf("Enter a non-negative decimal number: ");
    if (scanf("%u", &n) != 1) {
        printf("Invalid input.\n");
        return 1;
    }
    printf("Binary representation: ");
    decimal_to_binary(n);
    printf("\n");
    return 0;
}
```

Compilation and Execution Commands

gcc hdoc68.c -o hdoc68

./hdoc68

Screenshot Spaces

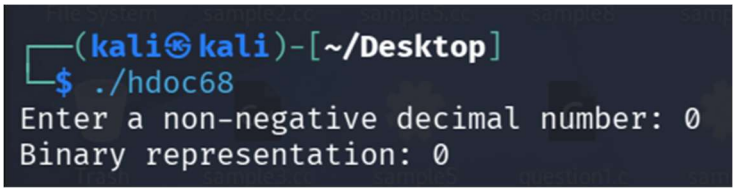
Compilation and Execution



Test Case 1

Input: 0 Expected output: Binary representation: 0

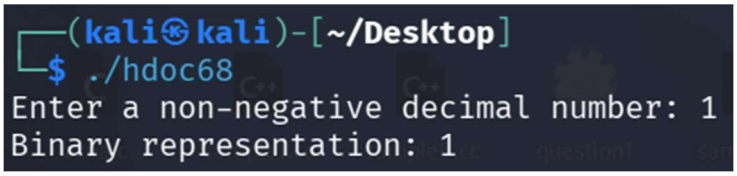
Purpose: Zero / base case



Test Case 2

Input: 1 Expected output: Binary representation: 1

Purpose: Single binary digit



Test Case 3

Input: 2 Expected output: Binary representation: 10

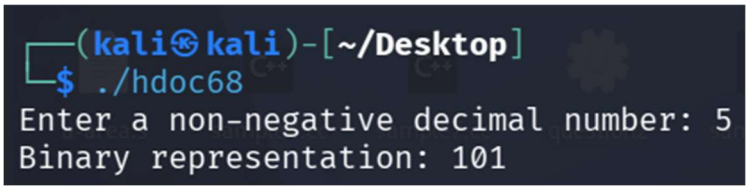
Purpose: Small even number



Test Case 4

Input: 5 Expected output: Binary representation: 101

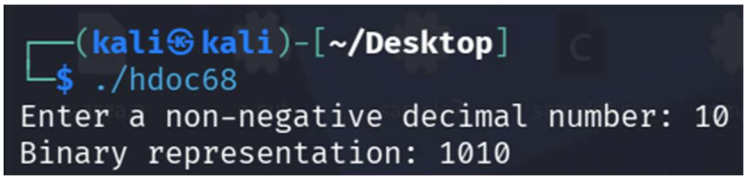
Purpose: Odd number



Test Case 5

Input: 10 Expected output: Binary representation: 1010

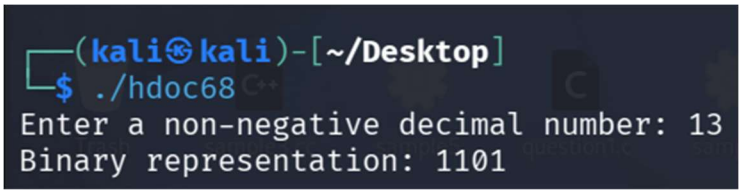
Purpose: Even number



Test Case 6

Input: 13 Expected output: Binary representation: 1101

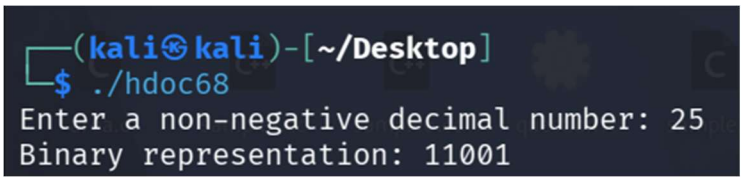
Purpose: Typical input



Test Case 7

Input: 25 Expected output: Binary representation: 11001

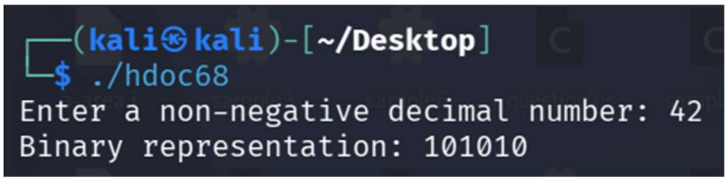
Purpose: Larger odd number



Test Case 8

Input: 42 Expected output: Binary representation: 101010

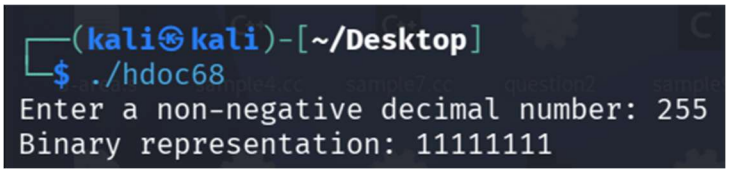
Purpose: Alternating digits



Test Case 9

Input: 255 Expected output: Binary representation: 11111111

Purpose: Eight one-bits



Test Case 10

Input: 1024 Expected output: Binary representation: 10000000000

Purpose: Power of two

